

Formulation of the relationship between temperature and groundwater conductivity in varying geological conditions

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Abstract: The Yogyakarta and Central Java regions of Indonesia have diverse geological conditions. This geological diversity has encouraged researchers to conduct hydrogeological research, particularly on the parameters of temperature (T) and electrical conductivity (EC) of groundwater. This research was carried out using hydrogeological mapping methods and T/EC measurement experiments both in the studio and in the laboratory. This study aims to formulate the relationship between T and EC of groundwater under various geological conditions. Geological variations, particularly lithology and geothermal processes, can influence the thermal dynamics and electrical properties of groundwater. The selected locations represent a range of geological conditions, including geothermal, karst, alluvial, and volcanic areas. Temperature and conductivity data were measured directly in the field at 10 wells and 33 springs, with support from laboratory tests. Additionally, experiments were conducted to determine the EC at various temperatures. The results showed that the relationship between T and EC in all samples is exponential in nature. This study developed a general formula based on a second-order polynomial graph, with varying coefficients A , B , and C . A wide range of coefficient values is shown by geothermal springs, while less variable values are shown by water in karst and volcanic areas. The formula obtained helps validate the EC value (σ) to be a valid field EC value (σ_{FV}), which is in accordance with local geological characteristics.

Keywords: alluvial group, electrical conductivity, geothermal group, karst, temperature, volcanic group

INTRODUCTION

Groundwater has been extensively studied recently due to its essential role as a water resource (Listyani, Prabowo and Suparta, 2023). Groundwater is a basic necessity for daily life. Human needs for groundwater must be met in terms of quantity and quality. The geological characteristics of an area determine groundwater quality (Listyani, 2022), and geological structures, such as faults, can also impact groundwater quality (Faryabi, 2025). Therefore, groundwater exploration must continue to be encouraged to determine its potential and prospects in an area (Ishola, 2025), including the use of geophysical methods (Ogunkoya, 2025). The electrical conductivity (EC) para-

meters have also been used to assess groundwater quality (Maciel Paula de *et al.*, 2024) and soil quality (Chen *et al.*, 2025).

Groundwater can be found in normal-temperature springs in various regions, as well as in hot springs in geothermal prospect areas. Multiple studies have been conducted to examine the geological link to groundwater potential, both in normal and hot springs. Hydrochemical methods have also been widely used in geothermal exploration activities (Alam *et al.*, 2019).

Hydrochemical studies have been conducted extensively by researchers, including various field measurements of chemical properties using digital tools (Setiawan *et al.*, 2020; Listyani, Prabowo and Suparta, 2023). Groundwater quality measurements using digital technology have been extensively conducted,

including the application of spectrophotometric methods to measure turbidity (Harini and Sulandari, 2013) and conductivity (Martanto *et al.*, 2013). Digital technology has been used in remote water quality monitoring and control systems (Merucahyo *et al.*, 2014).

In addition to chemical properties, the physical properties of groundwater are also crucial in determining its quality. One important groundwater physical property parameter to know is the *EC* value. Measurements of *EC* are usually valid at a temperature (*T*) of 25°C, so the *EC* temperature coefficient is typically compensated to 25°C when reporting (Nicholson, 1993). Therefore, a correction factor is needed to ensure that the *EC* values measured with standard equipment reflect the actual values.

Groundwater conductivity is influenced by various factors, including temperature, porosity, mineral composition, and local geological structure (Li, Zhao and Yang, 2020). Groundwater temperature plays a crucial role in the chemical and physical processes occurring within soil and rock layers, where temperature changes can increase the rate of chemical reactions and alter the *EC* characteristics (Zheng *et al.*, 2024). In altered rock formations, such as volcanic or sedimentary rocks rich in clay minerals, groundwater *EC* can increase with increasing *T* due to the increased amount of dissolved ions (Wang, 2020). Conversely, in hard rocks with low porosity, the relationship between *T* and *EC* can be more complex, highly dependent on pore structure and the presence of fractures (Xu *et al.*, 2015).

Several previous studies have shown that groundwater flow rate and aquifer depth also influence the relationship between *T* and groundwater *EC*. Lekshmy and Achari (2026) stated that increasing *T* at greater depths can cause significant changes in groundwater *EC* properties, which in turn affect the sustainability and efficiency of geothermal systems. Modelling the relationship between *T* and groundwater *EC* is crucial, both for groundwater resource management planning and for geothermal energy exploration applications (Doost *et al.*, 2026).

This study aims to establish the relationship between *T* and *EC* of groundwater under varying geological conditions by observing the pattern of this relationship. The formulation of the relationship between these parameters will help improve data validation. In addition, this study also obtained valid *EC* values according to the formula found. The results of this study are expected to contribute to the development of mathematical models that can predict groundwater *EC* based on *T* data and local geological parameters. The relationship between *T* and *EC* of groundwater is crucial in understanding groundwater flow and thermal properties in various geological formations. A better understanding of the relationship between *T* and *EC* will also aid in designing a more efficient groundwater quality monitoring system, taking into account the peculiarities of the geological conditions.

MATERIALS AND METHODS

GEOLOGY OF THE RESEARCH AREA

Several areas within the provinces of Yogyakarta and Central Java, Indonesia, were selected for this study based on their geological characteristics. These areas include: (a) geothermal prospect

areas, namely Guci and Ungaran; (b) limestone areas, namely the Gunungsewu Karst; (c) volcanic areas, namely the slopes of Mount Merapi; and (d) alluvial areas in Parangtritis and its surroundings. The geology of the study areas is briefly described as follows.

Geothermal areas are usually characterised by hot springs, which are the object of this study, including the Ungaran and Guci areas. Mount Ungaran is a Quaternary volcano composed of lava, volcanic breccia, tuff, and shallow intrusions associated with hydrothermal systems. This area exhibits surface manifestations, including hot springs, fumaroles, and hydrothermal alteration, indicating the presence of a water-dominated geothermal system. The hot water in this area has a temperature of 40–70°C, with varying components, reflecting the mixing process between meteoric water and hydrothermal fluids (Arnórsson, 2000).

The Guci geothermal area is located on the southwest slope of Mt. Slamet, Central Java, and is composed of andesitic-basaltic lava rocks, pyroclastic breccias, tuffs, and lahar deposits. The rocks of the Guci area are dominated by relatively fresh and permeable young Quaternary volcanic products as well as lahar and alluvial deposits at the foot of the mountain (Hochstein and Sudarman, 2008). The hot springs have a surface temperature of 47–70°C. The hot water is of the bicarbonate-sulphate type, indicating a water-dominated geothermal system with a mixture of meteoric and magmatic water (Arnórsson, 2000).

The Gunungsewu karst area is situated in the southern part of Java Island, with its constituent rocks dominated by the Wonosari Formation, which comprises reef limestone, clastic limestone, and layered limestone. Typical landforms encountered include conical hills with a relative height of 30–60 m, with a kegelkarst pattern, dolines (sinkholes), uvalas, caves, and underground river systems. Gunungsewu landscape is a humid tropical karst, resulting from the interaction of geological, climatic, and hydrological factors (Ford and Williams, 2007).

The volcanic group of this research is represented by Mt. Merapi area. Mt. Merapi is one of the world's most active volcanoes, producing Quaternary volcanic rocks, including andesite-basaltic lava, pyroclastic breccia, and tuff. The stratigraphy of this region includes rocks from the Old Merapi, Middle Merapi, Young Merapi, and Modern Merapi (Newhall *et al.*, 2000; Camus *et al.*, 2000). Mount Merapi's magma has a high volatile content (H_2O , CO_2 , SO_2) (Costa *et al.*, 2013), making this mountain explosive and forming pyroclastic density currents. In the field, springs are often found in breccia aquifers or volcanic sandstone.

The alluvial group is represented by South Coast plain. The South Coast (Parangtritis) region is located in the southern part of the Yogyakarta Basin, directly bordering the Indian Ocean. The dominant rocks are young Quaternary sediments that fill the Yogyakarta-Bantul Graben, a product of the Merapi volcano and the Opak River system. The activity of the Progo River influences several alluvial areas on the South Coast of Yogyakarta.

RESEARCH MATERIALS

This research was conducted in several locations with varying geological conditions, including geothermal prospect areas (Ungaran and Guci), volcanic rock formations (Merapi), sedimentary rocks (Wonosari Formation), and alluvial deposits (South Coast) – Figure 1. The research locations were selected

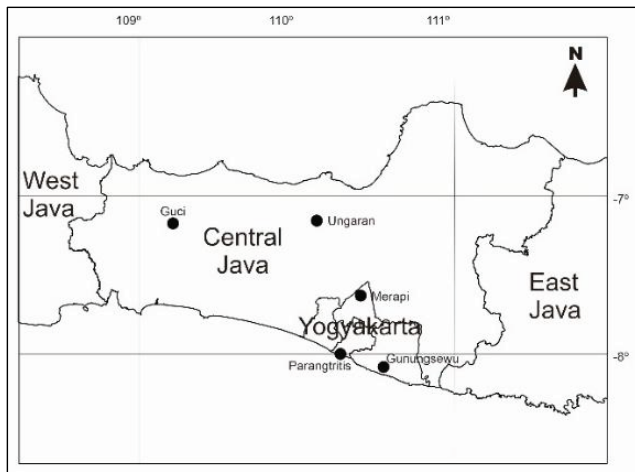


Fig. 1. Research location; source: own elaboration

based on differences in the constituent rocks and variations in T in groundwater sources, allowing for the study of the relationship between T and EC in the groundwater.

The object of study in this research is groundwater, obtained from wells, springs, and dolines. Dolines were chosen as the research object because, although dolines are lakes that collect surface water, in the Gunungsewu Karst area, these lakes are often supplied by groundwater (Listyani and Isnawan, 2021). The observation locations and the T and EC values of the groundwater sources measured directly in the field are presented in Table S1.

STUDY METHODS

Data collection

The research began with a literature review, followed by a field survey and data analysis (Fig. 2). The hydrogeological study in the field was supplemented by groundwater sampling. To observe variations in T and EC values across various rock types, groundwater source observations were conducted at 43 observation locations (Tab. S1). Several groundwater samples were analysed at selected locations in the laboratory and studio. Sample selection was based on rock variations and field measurements of T and EC .

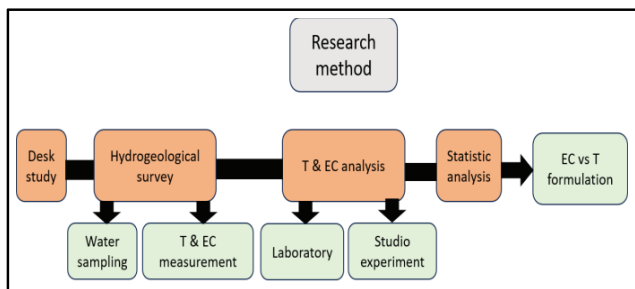


Fig. 2. Research method; T = temperature, EC = electrical conductivity; source: own elaboration

Field survey and geological characterisation

Geological characterisation is primarily performed to determine rock types based on field descriptions and petrological analyses of rock samples in the laboratory. These geological characteristics are then analysed to determine the relationship between groundwater T and EC .

• Measurement of T and EC in the field

Groundwater T measurements were conducted using a digital thermometer directly in dug wells or springs (Photo 1). Groundwater EC values were measured using Hanna and Eutech EC -meters calibrated with standard solutions. These field measurements were conducted at least 10 times at each location, and then the average value was determined.



Photo 1. Field and studio experiments to obtain electrical conductivity (EC) values at various temperatures (phot.: B. Simarmata)

• Rock and groundwater sampling

Rock samples were taken for petrological description of the rocks from the springs or around the wells. Groundwater samples were collected using polyethylene bottles for further analysis.

• Laboratory analysis

To further verify the field data, 20 groundwater samples were sent to an accredited laboratory for analysis of T and EC . To protect the samples from temperature changes before they were delivered to the lab, ice boxes are used. The results of the T and EC measurements in the laboratory are presented in Table S2.

• Studio experiment

Of the twenty samples tested in the water laboratory above, all samples were tested to analyse the relationship between T and EC at various temperatures. To achieve a higher T variation, heating was performed using a gas stove. Meanwhile, lower temperatures were obtained by cooling using a freezer in the studio or an ice box when in the field (Photo 1).

Data analysis

Of the 20 samples tested, only 18 could be further analysed. This analysis included a correlation analysis of the T and EC relationship (symbol σ) as well as mathematical modelling (Fig. 3). Formulations for the T and EC relationship were developed for each geological formation. Furthermore, based on the data, mathematical formulas were synthesised to understand the geological characteristics. Several steps in the analysis process can be detailed as follows.

• Statistical analysis

Statistical analysis was conducted to find the relationship pattern between T and EC based on the results of the studio sample experiment, namely T_E and σ_E . The results of this statistical analysis of the relationship between T_E and σ_E will produce a mathematical equation.

• Verification of experimental data with laboratory data

The equations obtained from the statistical analysis graphs were then verified with laboratory data. Verification was performed by calculating the EC value at laboratory temperature (σ_L) and calculating the error. If the error was small ($\leq 5\%$), then the σ_E and σ_L values were relatively similar, meaning there was no significant difference. Therefore, the σ_E value was used for the

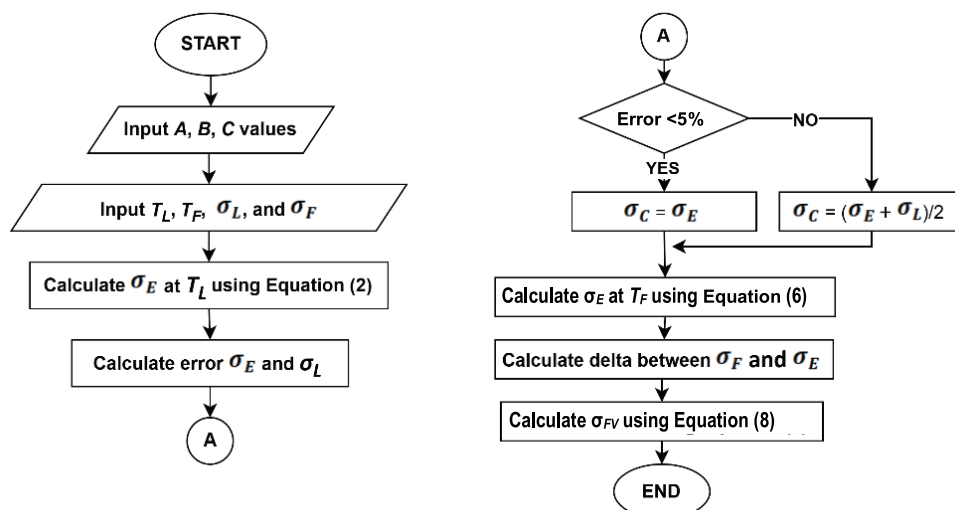


Fig. 3. Flow chart of the data analysis process; source: own elaboration

following process. However, if the error exceeded 5%, the EC value used next was the average of σ_E and σ_L .

• Field EC value validation based on mathematical equations

To obtain a valid field EC value (σ_F), this value needs to be validated as σ_{FV} . The σ_{FV} value is obtained from the following mathematical equation.

RESULTS AND DISCUSSION

CATEGORISATION OF GEOLOGICAL GROUP

The results of groundwater electrical conductivity (EC) measurements studied at various temperatures yielded varying relationships for each geological characteristic. Based on the geology of the studied area, the research areas can be divided into four groups:

- 1) geothermal group: includes the Ungaran, Guci, and Parangwedang areas;
- 2) karst group: covers the Gunungsewu area;
- 3) alluvial group: includes coastal and river alluvial around Parangtritis, Bantul;
- 4) volcanic group: consists of the Merapi and Sidomukti areas on the slopes of Mount Ungaran.

THE ELECTRICAL CONDUCTIVITY MEASUREMENT RESULTS

Given the importance of groundwater quality studies for human survival, precise methods and measurements are essential for assessing groundwater quality, especially in EC . The EC measurements at varying temperatures will result in inaccurate EC values. Therefore, temperature studies are necessary to obtain valid and accurate EC values. Measurement errors caused by temperature fluctuations can be corrected with temperature compensation (Wei *et al.*, 2010). The formulation of the relationship between EC and T , which is unique to specific geological conditions, is a novel aspect of this study.

Next, the EC test results at various temperatures of the studied water were statistically analysed to obtain an appropriate formula. These measurements, with an average error of $\leq 5\%$, were considered valid.

TEMPERATURE VS ELECTRICAL CONDUCTIVITY FORMULATION

Nonlinear regression analysis was employed to assess the correlation between T and EC of groundwater under various geological conditions. The graph obtained from analysing all samples showed a second-order polynomial trendline pattern. An example graph for each geological group is shown in Figure S1. These graphs will provide a differentiating picture of the T and EC relationships of the four geological groups.

The relationship equation between T and EC for each spring as well as the coefficient of determination (R^2) for each sample are shown in Table S3. Overall, the modelling results show an average R^2 of 0.9583. This means that the average sample has 95.83% of the dependent variable that can be explained by the independent variables in the model. In fact, there is a model that is close to the sample, namely the data from the Geyong Cave spring (E4). In this model, the R^2 value is 0.9917, close to 1. This means that the mathematical model in the form of a second-order polynomial successfully models the data well. The values of coefficients A , B , and C vary, as shown in Table S4.

From the graph of the relationship between T and EC , as shown in Figure S1, a general formula can be derived (Eq. (1)).

$$\sigma_T = AT^2 - BT + C \quad (1)$$

where: T = temperature ($^{\circ}\text{C}$), σ = electrical conductivity (EC) ($\mu\text{S}\cdot\text{cm}^{-1}$), σ_T = electrical conductivity (EC) at temperature T , and the A , B , and C = coefficients of the experimental results equation.

To get a validated EC value, the following steps are:

1. Calculate the experimental EC value (σ_E) at laboratory temperature (T_L) using Equation (2):

$$\sigma_E = AT_L^2 - BT_L + C \quad (2)$$

where: σ_E = experimental conductivity, and T_L = laboratory temperature.

2. Calculating the error of EC (Err.) between the experimental and laboratory EC :

$$Err. = \frac{\sigma_L - \sigma_E}{\sigma_L} 100 \quad (3)$$

If $Err. \leq 5\%$ then

$$\sigma_C = \sigma_E \quad (4)$$

If $Err. > 5\%$ then

$$\sigma_C = \frac{\sigma_E + \sigma_L}{2} \quad (5)$$

where: σ_C = calibrated conductivity at laboratory temperature.

3. Calculate the experimental EC value (σ_E) at field temperature (T_L) using Equation (6):

$$\sigma_E = \sigma_C + A(T_F^2 - T_C^2) - B(T_F - T_C) \quad (6)$$

Equation (6) is obtained as follows. Based on Equation (1), the field and calibrated conductivity values are as follows:

$$\sigma_E = AT_F^2 - BT_F + C \quad (6a)$$

$$\sigma_C = AT_C^2 - BT_C + C \quad (6b)$$

$$\sigma_E - \sigma_C = A(T_F^2 - T_C^2) - B(T_F - T_C) \quad (6c)$$

so the Equation could be transformed to get the experimental EC :

$$\sigma_E = \sigma_C + A(T_F^2 - T_C^2) - B(T_F - T_C)$$

where: T_F = the temperature in the field.

4. Calculating the difference (Δ) between the field EC and the experimental EC value (σ_E) at field temperature (T_L) between the experimental EC value (σ_E) at field temperature (T_L) and the field EC (σ_F) (see step 3):

$$\Delta\sigma = \sigma_E - \sigma_F \quad (7)$$

Searching for validated field EC (σ_{FV})

$$\sigma_{FV} = \sigma_F + \Delta\sigma \quad (8)$$

The final validated EC results from all observation locations are presented in Table S5.

Furthermore, the coefficients A , B , and C of the obtained formula have been analysed by adjusting the geological characteristics. The results can be summarised in Table S6 and Figure S2. For this grouping, the cold spring A5 on the slopes of Mt. Ungaran is included as a volcanic spring that is not affected by geothermal energy. Meanwhile, the warm spring in Parangwedang (C11), although located in an alluvial area, is included as a geothermally controlled spring.

Each geological group has distinct characteristics, where groundwater associated with geothermal has relatively large coefficients A , B , and C (see Tab. S6). This suggests that geothermal activity leads to significant variations in the validation of EC measurements. The values that must be corrected for geothermally controlled springs are very wide, depending on local geological conditions, of course. The lowest values for coefficients A , B , and C are found in groundwater from the alluvial and karst groups. In this group, the A , B , and C values have a relatively

narrow range, meaning that almost all springs found in karst and volcanic geological conditions have similar values, so that the correction of EC values will produce figures that are not significantly different.

CONCLUSIONS

Several springs have been studied that represent their respective geological conditions, namely springs in geothermal, karst, alluvial, and volcanic areas. Field surveys were conducted in the Mount Merapi area (volcanic springs), the South Coast area and its surroundings (springs/wells in alluvial areas), the Ungaran and Guci areas (dominated by geothermal springs), and the Gunungsewu area (a source of karst water). Measurements of temperature (T) and electrical conductivity (EC) were conducted to investigate the relationship between the two variables and to develop a generally applicable formula for this relationship. The results of the study showed an exponential relationship between T and EC of the groundwater studied.

The obtained formula reveals that geothermal-associated springs exhibit a wide range of coefficients, suggesting that the correction factors for groundwater in this area vary significantly. The narrow range of coefficient values is found in karst and volcanic groundwater, indicating that the EC values in these areas do not exhibit a wide range, or in other words, the relationship between T and EC of volcanic groundwater has many similarities. The resulting general formula helps correct EC values according to geological characteristics. Therefore, in addition to differences in coefficient values for various geological groups, this study also provides validated EC values.

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SUPPLEMENTARY MATERIAL

Supplementary material to this article can be found online at: https://www.jwld.pl/files/Supplementary_material_70_Listyani.pdf.

CONFLICT OF INTERESTS

All authors declare that they have no conflicts of interests.

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